

A Credible-Subpopulation Approach to Parallel Trends

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Abstract

Difference-in-differences (DiD) practitioners routinely confront the possibility that the parallel trends assumption fails for some treated cohorts but not others. The dominant responses keep the average treatment effect on the treated (ATT) as the target: heterogeneity-robust estimators point-identify it under parallel trends, while sensitivity methods such as HonestDiD report a set for it under bounded violations. Both inherit the problem that the ATT may be either biased or only weakly set-identified when some cohorts violate parallel trends. We propose a different response: change the estimand. We define a *credible-subpopulation local ATT* (LATT), the treatment effect for the subpopulation of cohorts whose parallel trends assumption is credible, and show it is point-identified precisely when the full ATT is not. The procedure is a reweighting of standard group-time effects toward credible cohorts, combined with honest sensitivity bounds on the residual violation of the selected set. We are explicit about the method’s scope: its advantage over the ATT grows with how informative pre-trends are about post-treatment violations and vanishes when they are uninformative, where only economic context can carry the identifying assumption. A reduced-form and a panel simulation study confirm that the LATT recovers a clean effect where the ATT estimator is biased, that the sensitivity bounds restore honest coverage where point estimates collapse, and that a relative-magnitudes restriction is self-undermining under uninformative pre-trends while a smoothness restriction is not.

1 Introduction

Difference-in-differences is among the most widely used research designs in empirical economics, and a large recent literature has clarified both its assumptions and its failure modes (Roth et al., 2023). The design’s credibility rests on the parallel trends assumption: absent treatment, the average outcomes of treated and comparison units would have evolved in parallel. In practice researchers are rarely certain this holds, and it has become standard to assess it by testing for pre-treatment differences in trends. As Roth et al. (2023) synthesize, this pre-testing practice is fragile: pre-trends tests are often underpowered against economically meaningful violations (Kahn-Lang and Lang, 2020; Roth, 2022), conditioning on passing them induces a pre-test selection bias (Roth, 2022), and even a clean pre-period does not guarantee a clean post-period (Kahn-Lang and Lang, 2020).

Two strands of methodology respond to this concern, and both retain the ATT as the estimand. The first develops heterogeneity-robust estimators for staggered adoption (Callaway and Sant’Anna,

2021; Sun and Abraham, 2021; Borusyak et al., 2024; de Chaisemartin and D’Haultfoeuille, 2020; Goodman-Bacon, 2021), which point-identify interpretable aggregations of group-time effects under a suitable parallel trends assumption but remain biased if that assumption fails. The second, exemplified by Rambachan and Roth (2023), abandons exact parallel trends and instead bounds how different post-treatment violations can be from the observed pre-trends, reporting a uniformly valid confidence *set* for the ATT. This is a substantial advance, but it inherits a structural limitation: when parallel trends fails for some cohorts, the ATT—an average over *all* treated cohorts, including the offending ones—is precisely the object that is hard to recover. The heterogeneity-robust point estimate is then biased, and the HonestDiD set can be wide and uninformative, because it must accommodate the worst offenders.

This paper takes a different route, familiar from other corners of causal inference: when the population target is not credibly identified, retreat to a subpopulation where it is. This is the logic of the local average treatment effect (Imbens and Angrist, 1994), of the optimal-subpopulation average treatment effect under limited overlap (Crump et al., 2009), and of overlap weighting (Li et al., 2018). In each case one trades the original estimand for a credibly-identified effect on a selected subpopulation, and defends the trade by characterizing whom the subpopulation comprises. We develop the staggered-DiD instance of this principle. We define the credible-subpopulation LATT—the weighted average of group-time effects restricted to cohorts whose parallel trends assumption is credible—and show that it is point-identified even when the full ATT is not, because the violations of the excluded cohorts do not enter it. Estimation is a transparent reweighting of the Callaway and Sant’Anna (2021) group-time effects; the credibility of a cohort is assessed from its pre-trends, so that the reweighting is a data-dependent selection, and we treat inference on the selected set with the sensitivity machinery of Rambachan and Roth (2023).

Our contribution is therefore primarily one of framing and estimand choice rather than of new estimation or inference technology, and we are deliberate about this. The method occupies a distinct point on a credibility–precision–scope frontier: relative to a heterogeneity-robust point estimate of the ATT it gives up scope (it targets a subpopulation) to buy back credibility; relative to a HonestDiD set for the ATT it gives up nothing in honesty but recovers a sharper, point-identified answer to a narrower question. We are equally explicit about the method’s limits. Its advantage is governed by how informative pre-trends are about post-treatment violations; when they are uninformative, selecting on them buys nothing, and the identifying assumption must be carried by economic context rather than by the data. We show, moreover, that the natural relative-magnitudes sensitivity restriction is *self-undermining* in exactly this regime—it anchors its bounds to the observed pre-trends and so reports false precision when those pre-trends are flat—whereas a smoothness restriction, which keys off structure rather than magnitude, is not.

The remainder of the paper is organized as follows. Section 2 fixes the staggered-DiD setup and the decomposition of event-study coefficients into a causal and a differential-trend component. Section 3 defines the credible-subpopulation LATT, states its identification and the properties of its estimator, and gives our recommended sensitivity-based inference together with an honest account

of what is provable and what is not. Section 4 reports the simulation study. Section 5 concludes.

2 Setup

We adopt the staggered adoption framework of Callaway and Sant’Anna (2021). There are periods $t = 1, \dots, T$ and units i indexed by their first treatment date $G_i \in \{2, \dots, T\} \cup \{\infty\}$, where $G_i = \infty$ denotes never-treated. Let $Y_{it}(g)$ denote the potential outcome in period t if unit i is first treated in period g , and $Y_{it}(\infty)$ the never-treated potential outcome. The building-block causal parameter is the group-time average treatment effect on the treated,

$$\text{ATT}(g, t) = \mathbb{E}[Y_{it}(g) - Y_{it}(\infty) \mid G_i = g]. \quad (1)$$

Summary parameters are weighted aggregations $\theta = \sum_g \sum_t w(g, t) \text{ATT}(g, t)$ with researcher-chosen weights, of which the overall ATT and the event-study path are special cases.

It is convenient to work with the event-study representation and the decomposition of Rambachan and Roth (2023). Let $\hat{\beta} = (\hat{\beta}'_{\text{pre}}, \hat{\beta}'_{\text{post}})'$ be an asymptotically normal vector of event-study coefficients estimated by any of the heterogeneity-robust procedures above, with $\sqrt{n}(\hat{\beta} - \beta) \rightarrow \mathcal{N}(0, \Sigma)$. The estimand decomposes as

$$\beta = \tau + \delta, \quad \tau_{\text{pre}} = 0, \quad (2)$$

where τ collects the causal effects (zero before treatment, by no anticipation) and δ is the differential trend between treated and comparison groups that would have occurred absent treatment. Parallel trends is the restriction $\delta_{\text{post}} = 0$, under which $\beta_{\text{post}} = \tau_{\text{post}}$; a pre-trends test is a test of $\delta_{\text{pre}} = 0$.

We will require the analogous objects at the cohort level. Write δ_g for the differential trend of cohort g and say that parallel trends *holds for* g if $\delta_{g, \text{post}} = 0$. A key feature of applications is that this may hold for some cohorts and fail for others: a policy adopted by cohort g may be as good as randomly timed while another cohort’s adoption coincides with a confounding shock.

Finally, we distinguish two regimes for how cohorts are judged credible, because they carry different inferential guarantees. Under *ex-ante* (or independent) selection, credibility is decided from pre-determined covariates, institutional knowledge, or an independent data split, so that the selected set is statistically independent of the estimation errors in $\hat{\beta}$. Under *data-driven* selection, credibility is assessed from the realized pre-trends $\hat{\beta}_{\text{pre}}$, so that the selected set depends on the estimation sample. The distinction is immaterial for the estimand and its identification, but central for inference, and we treat it explicitly in Section 3.

3 The credible-subpopulation LATT: approach and recommendations

3.1 Estimand and identification

Let S denote a set of cohorts judged to have credible parallel trends, produced by a selection rule R that we specify below. We define the target as the treatment effect for that subpopulation.

Definition 1 (Credible-subpopulation LATT). *For cohort weights $w_g > 0$ (e.g. proportional to cohort size), the credible-subpopulation local ATT is*

$$\theta_S = \frac{\sum_{g \in S} w_g \text{ATT}(g, \cdot)}{\sum_{g \in S} w_g}, \quad (3)$$

where $\text{ATT}(g, \cdot)$ is any within-cohort aggregation of interest (a fixed event-time, or an average over post-treatment periods).

The value of the target is that it is identified under a weaker condition than the ATT.

Proposition 1 (Identification). *If parallel trends holds for every $g \in S$, i.e. $\delta_{g,\text{post}} = 0$ for all $g \in S$, then θ_S is point-identified by the reweighted group-time effects, $\theta_S = (\sum_{g \in S} w_g \beta_{g,\text{post}}) / \sum_{g \in S} w_g$, and this holds irrespective of whether $\delta_{g,\text{post}} \neq 0$ for cohorts $g \notin S$.*

Proof. For $g \in S$, $\beta_{g,\text{post}} = \tau_{g,\text{post}} + \delta_{g,\text{post}} = \tau_{g,\text{post}}$ by the hypothesis and (2). The reweighted sum over S therefore equals the corresponding weighted average of causal effects, which is θ_S by definition. Cohorts outside S receive zero weight, so their differential trends do not enter. $\square$

Proposition 1 is elementary, but it is the paper’s central point: by *changing the target* to the credible subpopulation, one converts a partially-identified problem (the ATT, when some cohorts violate parallel trends) into a point-identified one, at the cost of answering a narrower question. This is exactly the trade made by Imbens and Angrist (1994), Crump et al. (2009), and Li et al. (2018) in other settings, and it carries the same interpretive obligation: one must characterize the subpopulation S that the estimand describes.

3.2 Estimation

The estimator $\hat{\theta}_S$ replaces population objects by their Callaway and Sant’Anna (2021) sample analogues and reweights over the selected cohorts. Its large-sample behavior is standard under ex-ante selection.

Proposition 2 (Consistency and efficiency). *Under ex-ante selection and the regularity conditions of Callaway and Sant’Anna (2021), $\hat{\theta}_S$ is consistent for θ_S . If parallel trends holds for all cohorts, $\hat{\theta}_S$ is consistent for the ATT, with an efficiency loss relative to estimators that use all cohorts.*

Under data-driven selection the selected set is random and correlated with the estimation errors, which introduces a finite-sample pre-test bias of the kind analyzed by Roth (2022).

Proposition 3 (Pre-test bias). *Under data-driven selection, $\hat{\theta}_S$ carries a finite-sample bias that vanishes as the per-cohort information grows and the selection rule becomes consistent for the credible set.*

We emphasize, consistent with Proposition 3 and with our simulations below, that the honest claim is consistency, not finite-sample unbiasedness: the data-driven estimator is close to, but not exactly on, the target in small samples, and the gap closes with sample size.

3.3 Honest inference on the selected set

Selecting credible cohorts does not by itself deliver parallel trends: a cohort with flat pre-trends may still have a nonzero post-treatment differential trend, and the selection removes gross offenders but not subtle residual violations. We therefore recommend accompanying the point estimate with the sensitivity analysis of Rambachan and Roth (2023), applied to the *selected-cohort aggregate* event study. Concretely, one imposes a restriction $\delta_{S,\text{post}} \in \Delta$ on the residual differential trend of the selected set and reports the implied confidence set for θ_S . Among the restrictions of Rambachan and Roth (2023), the smoothness class $\Delta^{SD}(M) = \{\delta : |\delta_{t+1} - 2\delta_t + \delta_{t-1}| \leq M \ \forall t\}$ admits the fixed-length confidence intervals (FLCIs) of Armstrong and Kolesár (2018), which are near-optimal for convex centrosymmetric restrictions and which we adopt.

The validity of this step depends on the selection regime. Under ex-ante selection the selected set is independent of $\hat{\beta}$, so the uniform validity results of Rambachan and Roth (2023) apply verbatim to the selected-cohort event study, and no new inference theory is required. Under data-driven selection the situation is more delicate: the selection is a function of $\hat{\beta}_{\text{pre}}$, and a classical impossibility result of Leeb and Pötscher (2005) precludes uniformly valid inference from the naive post-selection estimator without conditioning, sample-splitting, or randomization. Our simulations (Section 4) find that in practice the data-driven procedure exhibits no systematic coverage distortion—full-data and split implementations cover almost identically—so that the naive procedure is typically valid; but this is a pointwise, not a worst-case uniform, statement, and we are explicit about the distinction. A researcher who requires a worst-case guarantee can obtain it cheaply by selecting on pre-determined information, by splitting the sample, or by randomized (data-carving) selection in the sense of Fithian et al. (2014), all of which restore the independence that Proposition 1’s inferential counterpart needs.

A further subtlety arises because the selected set is random. The FLCI at bound M is valid for a given selected set only if that set’s residual differential trend lies in $\Delta^{SD}(M)$; since the set is random, its residual curvature is a random variable, and unconditional coverage requires M to dominate that variable with high probability.

Lemma 1 (Calibration to the random selected set). *Under data-driven selection, choosing M equal to the mean residual curvature of the selected aggregate undercovers; unconditional coverage requires M to exceed the mean and to approach an upper quantile of the residual-curvature distribution, with the FLCI’s sampling slack providing a partial buffer.*

Lemma 1 is confirmed sharply in Section 4 and is a practical warning: the sensitivity bound must be set conservatively relative to the *typical* selected set, not merely to satisfy it on average.

Finally, we caution against one otherwise-natural choice of Δ . The relative-magnitudes restriction $\Delta^{RM}(\bar{M})$ of Rambachan and Roth (2023), following Manski and Pepper (2018), bounds post-treatment violations by $\bar{M}$ times the observed maximal pre-treatment violation. Because it anchors its bound to the observed pre-trends, it is self-undermining precisely in the regime where our method’s value is in question.

Proposition 4 (Relative magnitudes is self-undermining). *The identified-set half-width under $\Delta^{RM}(\bar{M})$ is proportional to the observed pre-trend magnitude. When pre-trends are uninformative about post-treatment violations, the observed pre-trends are small while the true violation need not be, so $\Delta^{RM}(\bar{M})$ produces intervals whose width tends to zero even as the bias does not, yielding undercoverage. A structural restriction such as $\Delta^{SD}(M)$ does not share this defect, since its bound keys off smoothness rather than pre-trend magnitude.*

3.4 Scope: when to prefer the LATT

The method’s advantage over the ATT is not uniform; it is governed by how informative pre-trends are about post-treatment violations. When they are informative, selecting on them isolates credible cohorts and the LATT recovers a clean effect while the ATT estimator remains biased. When they are uninformative, selection is uninformative about the post-period, the residual violation of the selected set is as large as for any set, and the method offers no advantage over the ATT—there is no free lunch, and the identifying assumption must be defended from economic context. We regard a formal decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT, in the sense of Armstrong and Kolesár (2018), as the principal open theoretical question; the simulations below map its empirical shadow.

4 Simulation study

We study the method in a reduced-form normal model, in which event-study coefficients are drawn directly from their asymptotic distribution, and in a full panel micro-simulation, in which individual data are generated and the coefficients and their covariance are estimated. The reduced-form model isolates the inferential mechanics, where the underlying theory is exact; the panel model confirms that the conclusions survive realistic estimation.

4.1 The estimand: recovering a clean effect where the ATT is biased

We first illustrate Proposition 1. Twelve cohorts share a common true effect of one; a subset violate parallel trends in the post-period, biasing any estimator of the ATT. A heterogeneity-robust estimator of the ATT and the reweighted LATT are compared to the truth. Table 1 reports the result. The ATT estimator is biased by +0.27, entirely on account of the offending cohorts, while

Table 1: The estimand demonstration (reduced-form model). True effect = 1 for all cohorts.

	ATT estimator	LATT (ours)
Point estimate	1.267	1.013
Bias vs. truth	+0.267	+0.013
<i>Bias as sampling noise s falls:</i>		
$s = 0.40$	+0.269	+0.048
$s = 0.20$	+0.267	+0.005
$s = 0.12$	+0.267	+0.000
$s = 0.03$	+0.267	−0.000

the LATT recovers the credible-subpopulation effect. The right panel sweeps the sampling precision: the LATT’s bias vanishes as precision grows—the pre-test bias of Proposition 3—whereas the ATT estimator’s bias is frozen, because it is a wrong-target problem, not a noise problem.

4.2 The scope condition

Figure 1 traces the method across the axis that governs it: the informativeness of pre-trends about post-treatment violations. The bias of the ATT estimator is flat—it never consults pre-trends—while the LATT’s advantage is the vertical gap between the two, zero when pre-trends are uninformative and maximal when they are informative. The lower-left panel shows that the causal coverage of a point-based interval collapses at low informativeness; crucially, this collapse is driven by the residual violation of the surviving cohorts (an identification gap), not by selection distortion, since sample-splitting—which removes selection distortion by construction—collapses identically. This motivates the sensitivity bounds of the next subsection, and confirms that when pre-trends are uninformative the method correctly reports no advantage.

4.3 Honest inference restores coverage

We validate the smoothness-class FLCI applied to the selected aggregate. Table 2 confirms the exact honest behavior: the FLCI covers when and only when the assumed bound M dominates the true residual curvature C , while both a naive point interval and a linear-extrapolation point interval collapse to zero coverage the instant any curvature is present. Figure 2 shows the same result across a continuum of curvature: the point estimates collapse as the confound curves, the FLCIs remain honest exactly up to their assumed M , the price of honesty is interval width set by M , and the sensitivity curve delivers an interpretable breakdown value. Because $\Delta^{SD}(M)$ keys off smoothness rather than pre-trend magnitude, this honesty does not degrade with informativeness—in contrast to the relative-magnitudes restriction of Proposition 4.

Master-axis sweep: everything is governed by pre-trend informativeness
($\theta=1$ for all cohorts, so true ATT = true LATT = 1)

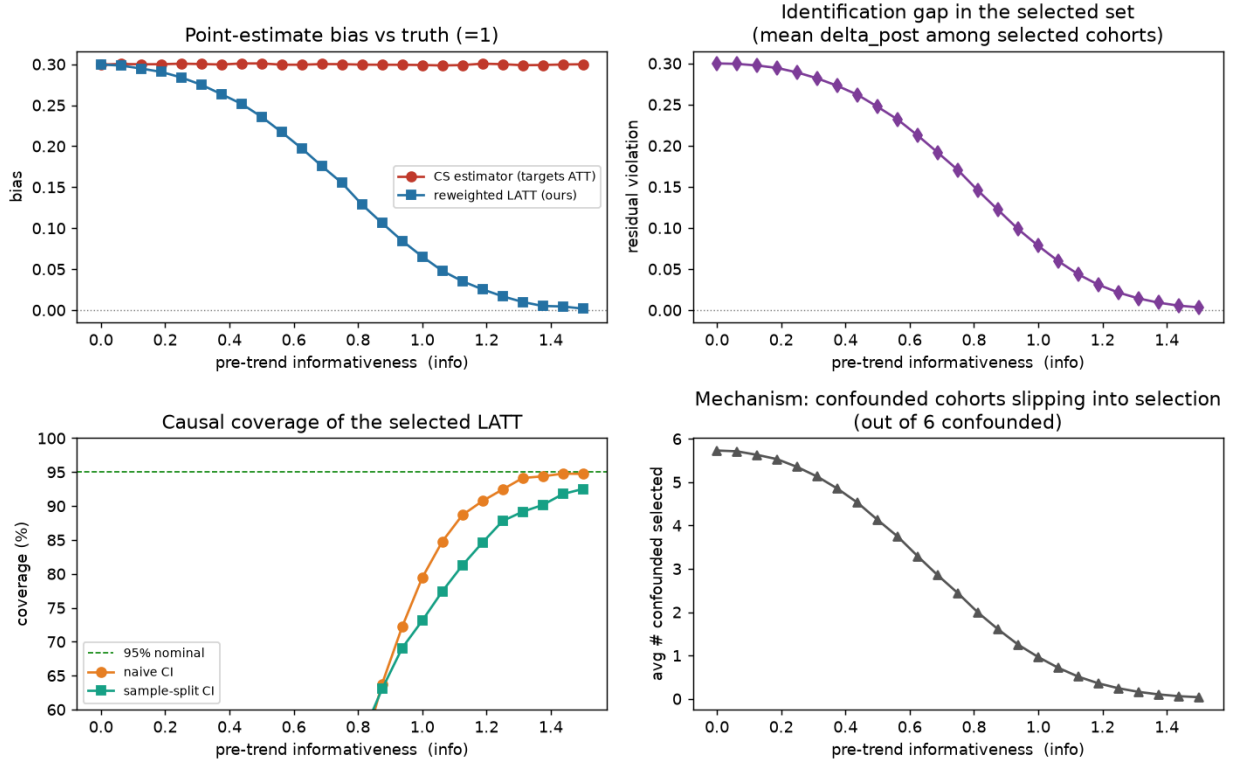


Figure 1: Scope condition. Everything is governed by pre-trend informativeness. The ATT estimator’s bias is constant; the LATT’s advantage is the gap between the lines. Causal-coverage collapse at low informativeness is an identification gap (surviving cohorts still violate parallel trends), not selection distortion—sample-splitting collapses identically.

4.4 Selection does not distort the bounds, but calibration must respect randomness

Two questions remain for the data-driven regime: does selection distort the FLCI, and how should M be chosen given that the selected set is random. Figure 3 answers both. Full-data selection and sample-splitting cross nominal coverage at the same bound, so selection introduces no systematic distortion of the FLCI—consistent with the pointwise reading of Leeb and Pötscher (2005) in favorable regimes, and confirming that splitting is not needed on this account. At the same time, a bound M set to the *mean* residual curvature of the selected set (0.031) undercovers, reaching only about 88%; nominal coverage is attained at $M \approx 0.039$, above the mean though below the strict maximum (0.050), as Lemma 1 anticipates. The practical recommendation is to calibrate M to the typical selected set, not to its average.

Table 2: FLCI coverage of the causal target (reduced-form model): valid iff $M \geq C$.

True curvature C	Assumed M	FLCI coverage	Point-estimate coverage
0.5	0.5 ($= C$)	94.7%	0.0%
0.5	1.0 ($> C$)	100%	0.0%
1.0	0.5 ($< C$)	0.0%	0.0%
1.0	1.0 ($= C$)	95.2%	0.0%

Full Layer 2 (validated): SD(M) FLCI restores honest coverage where point estimates collapse

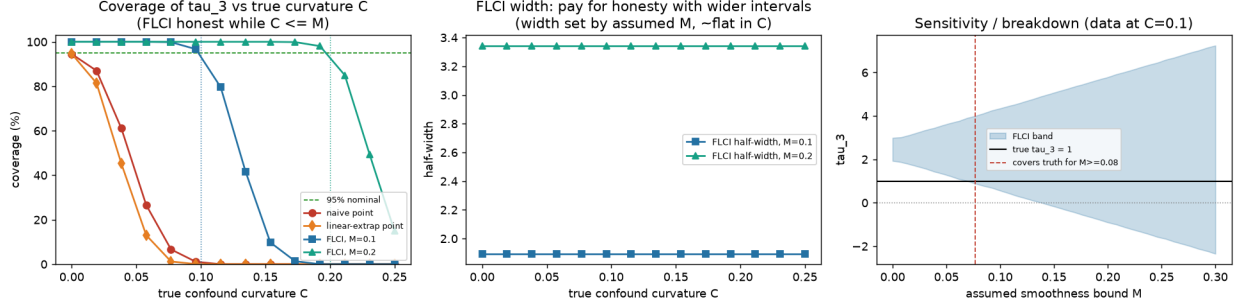


Figure 2: Sensitivity bounds restore honest coverage. Point estimates collapse as the confound curves; the FLCI is honest exactly while $M \geq C$; interval width is the price of honesty, set by M ; the right panel is the sensitivity/breakdown curve.

4.5 Panel confirmation

The reduced-form conclusions survive the full panel micro-simulation, in which individual outcomes are generated for a staggered design, cohort-level pre- and post-treatment coefficients are estimated against a common never-treated control, and the covariance is estimated from the data. The estimand demonstration is unchanged—the ATT estimator’s bias is flat near +0.30 while the LATTT’s bias falls toward zero with precision—and the estimated covariance is well calibrated where selection is deterministic. The residual gap at intermediate informativeness reflects the selection-induced component of variance that a conditional standard error omits, an independent confirmation of the inferential issues discussed above.

5 Conclusion

We have argued that when parallel trends fails for some treated cohorts, a productive response is to change the estimand rather than to defend or bound the ATT. The credible-subpopulation LATTT is point-identified precisely when the ATT is not, is estimated by a transparent reweighting of standard group-time effects, and can be accompanied by honest sensitivity bounds on the residual violation of the selected set. The approach is a specific instance of a general principle—retreat to a credibly-identified subpopulation—already established in the analysis of instrumental variables and of limited overlap.

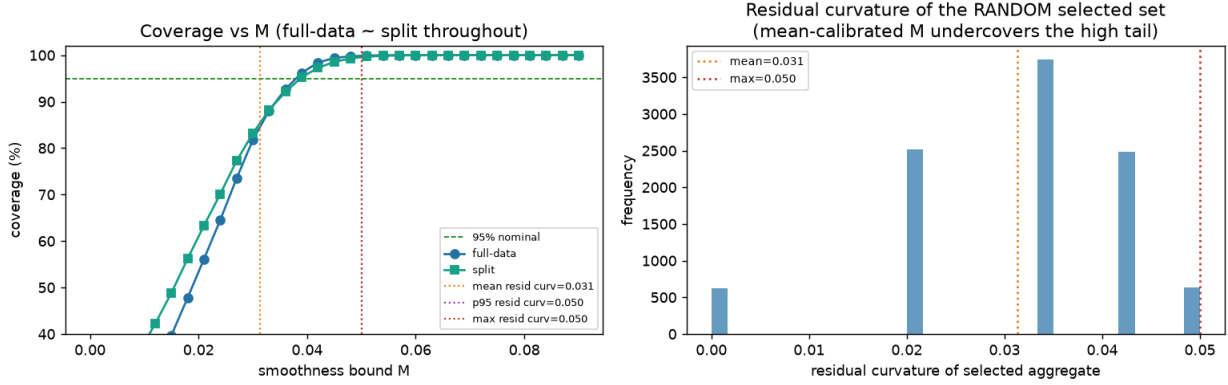


Figure 3: Calibrating the sensitivity bound to the random selected set. Full-data and split coverage cross nominal at the same M (no selection distortion); a mean-calibrated M undercovers, with nominal coverage reached above the mean residual curvature.

We have been explicit about the method’s advantages and its weaknesses. Its advantage over the ATT is that it delivers a sharp, credibly-identified answer where the ATT estimator is biased and the ATT sensitivity set is wide; its transparency about which comparisons identify the effect is inherited from Callaway and Sant’Anna (2021). Its weaknesses are equally concrete. The estimand is a subpopulation effect and must be interpreted as such, with the composition of the credible set described rather than assumed away. Its advantage is contingent on pre-trends being informative about post-treatment violations, and when they are not, only economic context can carry the identifying assumption. Under data-driven selection the estimator carries a vanishing pre-test bias, and worst-case uniform inference is precluded without ex-ante selection, sample-splitting, or randomization—though we find no systematic distortion in practice. The sensitivity bound must be calibrated to the typical selected set, and the relative-magnitudes restriction should be avoided in favor of a structural one, since the former is self-undermining under uninformative pre-trends.

The principal open question is theoretical: a decision-theoretic characterization of the region in which the point-identified LATT dominates the set-valued ATT would convert the empirical scope condition documented here into a formal recommendation, and would sharpen the choice between the two responses to imperfect parallel trends. We regard the framing developed here—estimand choice as the lever, with credibility purchased by narrowing the target—as the paper’s contribution, and the accompanying honesty about scope as inseparable from it.

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